

ESP32 3 — Wi-Fi Connection and Network Information

Tutorial Group: ESP32

Goal

In this tutorial, you will use the ESP32's built-in Wi-Fi capability to connect to a wireless network. Once connected, the ESP32 will display the team's name, ESP32 IP address, and ESP32 MAC address on the OLED.

This tutorial builds on *ESP32 I2C_OLED_Display*, so the OLED and I²C portions are not repeated here. The ESP32's Wi-Fi hardware is built into the microcontroller; no additional Wi-Fi shield or module is required.

What You Need

- ESP32-WROOM-32
- USB cable
- Computer with the Arduino IDE and ESP32 support installed
- Wi-Fi network with an SSID and password
- Breadboard and jumper wires, if needed
- OLED display with I²C interface

Part 1 — Wi-Fi and the ESP32

The ESP32 contains built-in Wi-Fi hardware. For this tutorial, the ESP32 will operate as a Wi-Fi station, sometimes called a client.



The access point provides the wireless connection and normally assigns the ESP32 an IP address. You will provide the ESP32 with the network's SSID and password.

Part 2 — Connect the OLED

Use the same OLED connections from *ESP32 I2C_OLED_Display*:

OLED	ESP32
VCC	3.30v
GND	GND
SDA	GPIO 21
SCL	GPIO 22

The OLED will display the network information after the ESP32 connects.

Part 3 — Define the Team and Wi-Fi Information

At the top of the program, define the team name and Wi-Fi credentials. For those *Hitchhiker's Guide to the Galaxy* fans who recognize the team name, remember: "Don't Panic".

```
const char teamName[] = "Team 42";

const char* ssid = "YourNetworkName";
const char* password = "YourPassword";
```

Replace the values with your team's information. The team name is a const char array, as in the previous tutorial. The SSID and password are also stored as character strings.

For testing, enter your own Wi-Fi network SSID and password in the program. During the competition, you will be provided with the network SSID and password to use.

Part 4 — Include the Wi-Fi Library

Add the ESP32 Wi-Fi library along with the OLED libraries:

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <WiFi.h>
```

The Wi-Fi library provides the functions needed to control the ESP32's built-in Wi-Fi hardware.

Part 5 — Connect to the Wi-Fi Network

Start the connection with:

```
WiFi.begin(ssid, password);
```

The ESP32 may take a short amount of time to establish the connection. We can wait until it succeeds:

```
while (WiFi.status() != WL_CONNECTED)
{
    Serial.print(".");
    delay(500);
}
```

The while loop is different from the loop() function. Here, waiting is intentional because the rest of the program cannot obtain a useful network address until the ESP32 has connected, and the Serial.print(".") gives us an indication the program is working and waiting on the connection to complete.

Part 6 — Obtain the IP Address

Once connected, obtain the IP address with:

```
WiFi.localIP()
```

For example, an access point might assign an address such as 192.168.1.37. The actual address depends on the network.

```
Serial.println(WiFi.localIP());
```

Part 7 — Obtain the MAC Address

Every network interface has a MAC address. The ESP32's Wi-Fi MAC address can be obtained with:

```
WiFi.macAddress()
```

A MAC address might look like 24:6F:28:12:34:56. Unlike the IP address, the MAC address identifies the network interface itself and normally does not change when the ESP32 connects to a different network.

Part 8 — Display the Information

Combine the Wi-Fi information with the OLED display. The following is a complete tutorial:

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <WiFi.h>

const char teamName[] = "Team 42";

const char* ssid = "YourNetworkName";
const char* password = "YourPassword";

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64

Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

void setup()
{
  Serial.begin(115200);

  // Initialize OLED
  if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C))
  {
    Serial.println("OLED not found");
    while (1);
  }

  // Connect to Wi-Fi
  WiFi.begin(ssid, password);

  while (WiFi.status() != WL_CONNECTED)
  {
    Serial.print(".");
    delay(500);
  }

  Serial.println(" Wi-Fi connected");

  // Display network information
  display.clearDisplay();
  display.setTextColor(SSD1306_WHITE);
  display.setTextSize(1);
```

```

display.setCursor(0, 0);
display.println(teamName);

display.setCursor(0, 20);
display.print("IP: ");
display.println(WiFi.localIP());

display.setCursor(0, 40);
display.print("MAC:");
display.println(WiFi.macAddress());

display.display();

Serial.print("Team: ");
Serial.println(teamName);

Serial.print("IP address: ");
Serial.println(WiFi.localIP());

Serial.print("MAC address: ");
Serial.println(WiFi.macAddress());
}

void loop()
{
    // All of the code for this tutorial occurs in setup().
}

```

Upload the program. The Serial Monitor will display dots while the ESP32 connects to Wi-Fi. Once connected, the team's name, IP address, and MAC address should appear on both the OLED and in the Serial Monitor.

Wi-Fi troubleshooting: make sure the ESP32-WROOM-32 and the computer are connected to the same Wi-Fi network. If the ESP32 connects but later cannot communicate with the computer, check the computer's firewall settings and allow the required application or network traffic when prompted.

Part 9 — Why Is loop() Empty?

This is an important difference from many of the previous tutorials. Normally, Arduino programs use loop() because they need to repeatedly perform a task.

```

void loop()
{
    readSensor();
    updateDisplay();
    checkButton();
}

```

In this tutorial, the task is simply to start the ESP32, connect to Wi-Fi, obtain the network information, and display it. Once those things have happened, there is nothing else that needs to be repeatedly done.

The Arduino framework still requires a loop() function, so we provide one containing only a comment:

```

void loop()
{
    // All of the code for this tutorial occurs in setup().
}

```

The ESP32 continues executing the framework's loop, but there is nothing for your program to do in it.

Part 10 — Observe the Serial Monitor

Open the Serial Monitor and set it to 115200 baud. You should see information similar to:

```
Team: Team 42
.... Wi-Fi connected
IP address: 192.168.1.37
MAC address: 24:6F:28:12:34:56
```

The values on your ESP32 will almost certainly be different. Compare the information displayed on the OLED with the information in the Serial Monitor.

Part 11 — Experiment

Experiment 1 — Change the Team Name

Change the `teamName` declaration to your actual team's name and verify that both the OLED and Serial Monitor display the new name.

Experiment 2 — Disconnect the Network

Turn off the Wi-Fi access point or enter an incorrect password. Observe what happens. The program remains inside the while statement because it is waiting for the connection to succeed. This is intentional for this simple tutorial.

Experiment 3 — Compare IP and MAC Addresses

Reconnect the ESP32 and observe its IP address. If another network is available, connect to it and compare the addresses. Does the IP address change? Does the MAC address change? Consider why.

What You Should Understand

- The ESP32 has built-in Wi-Fi capability.
- The ESP32 can connect to a Wi-Fi access point as a client.
- An SSID identifies a wireless network.
- A password is used to authenticate with the network.
- An IP address identifies the ESP32 on the network.
- A MAC address identifies the ESP32's network interface.
- The IP address is the address other devices use to communicate with the ESP32 on the current network; the MAC address identifies the ESP32's network interface.
- Not every Arduino program needs useful code in `loop()`.
- `setup()` can perform a task once and then leave the processor with nothing further to do.
- `WiFi.begin()` starts the connection process.
- `WiFi.status()` can be used to determine whether the connection has succeeded.
- `WiFi.localIP()` obtains the assigned IP address.
- `WiFi.macAddress()` obtains the Wi-Fi MAC address

Before Moving On

You should be able to answer **yes** to all of the following:

- ☐ I can configure the ESP32 with an SSID and password.
- ☐ I can connect the ESP32 to a Wi-Fi network.
- ☐ I can determine the ESP32's IP address.

- ☐ I can determine the ESP32's MAC address.
- ☐ I can display network information on the OLED.
- ☐ I can print network information to the Serial Monitor.
- ☐ I understand the difference between an IP address and a MAC address.
- ☐ I understand why this tutorial does not need to perform any work in `loop()`.
- ☐ I understand that `setup()` runs once when the ESP32 starts.

Once this tutorial is complete, you should be comfortable connecting the ESP32 to Wi-Fi and using the OLED to display information about its network connection.